Practical 09: AlertDialog and Custom DialogFragment

Source: MODULE 2.pdf | Source pages: 7–14

Aim

To develop and test an Android application demonstrating alertdialog and custom dialogfragment.

Steps to Perform the Practical

1. Create an Android Studio project named DialogDemo using Empty Views Activity.

2. In activity_main.xml, add the Show AlertDialog and Show Custom Dialog buttons.

3. Create res/layout/dialog_custom.xml with a heading and an EditText for the user's name.

4. Create CustomDialogFragment.java and inflate dialog_custom.xml inside the dialog.

5. In MainActivity.java, show an AlertDialog with YES and NO buttons and open CustomDialogFragment when its button is tapped.

6. Run the app. Test both buttons, enter a name in the custom dialog, press Submit, and test Cancel.

Result

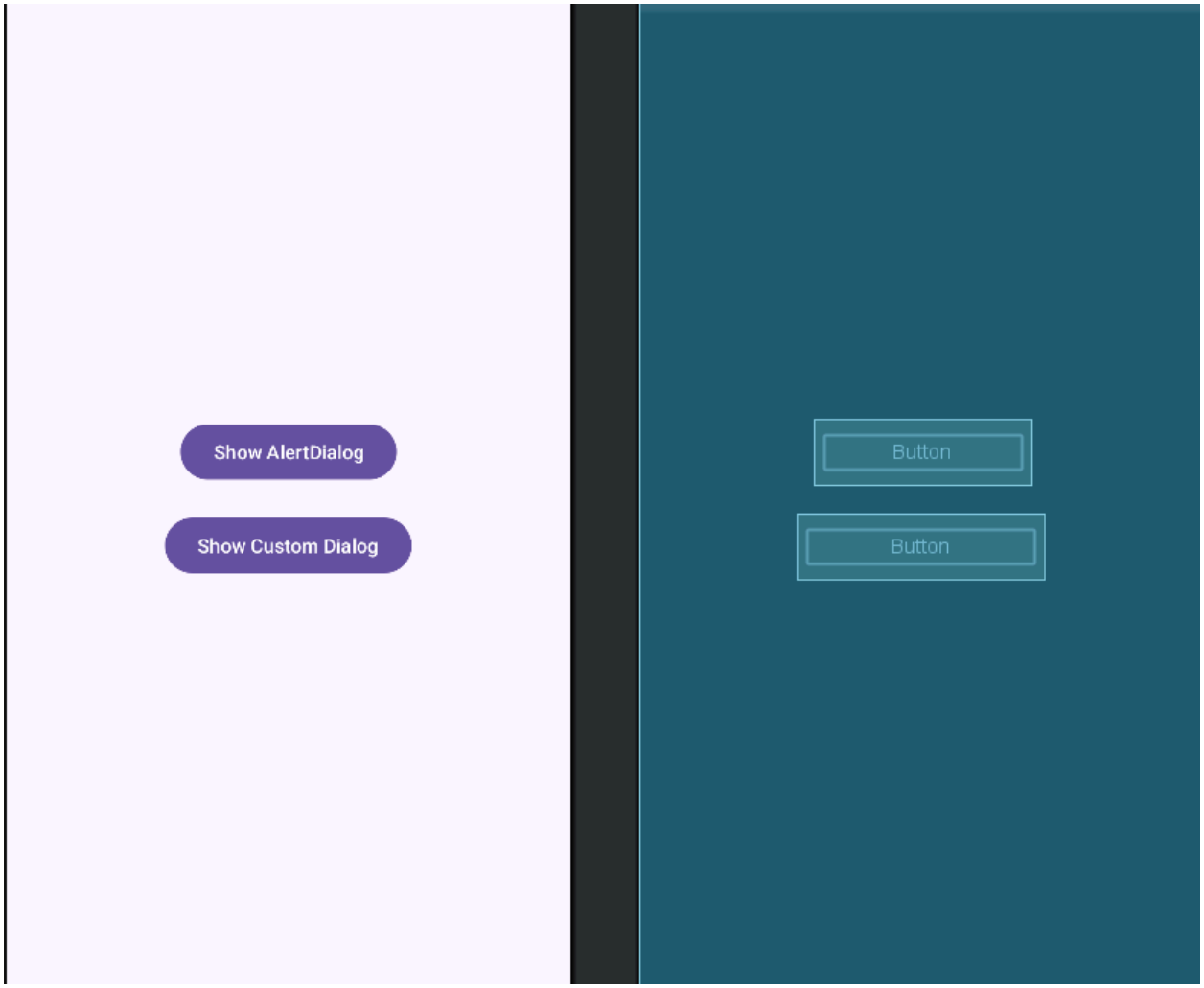
The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

2. Develop an Android application demonstrating AlertDialog and custom DialogFragment.

ProjectName: DialogDemo

File Name: activity_main.xml




```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:id="@+id/main"

   android:layout_width="match_parent" android:layout_height="match_parent"

   android:orientation="vertical"
```

```
   android:gravity="center" android:padding="20dp">

   <Button

        android:id="@+id/btnAlert"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Show AlertDialog"/>

   <Button

        android:id="@+id/btnCustom" android:layout_width="wrap_content"

        android:layout_height="wrap_content" android:text="Show Custom Dialog"

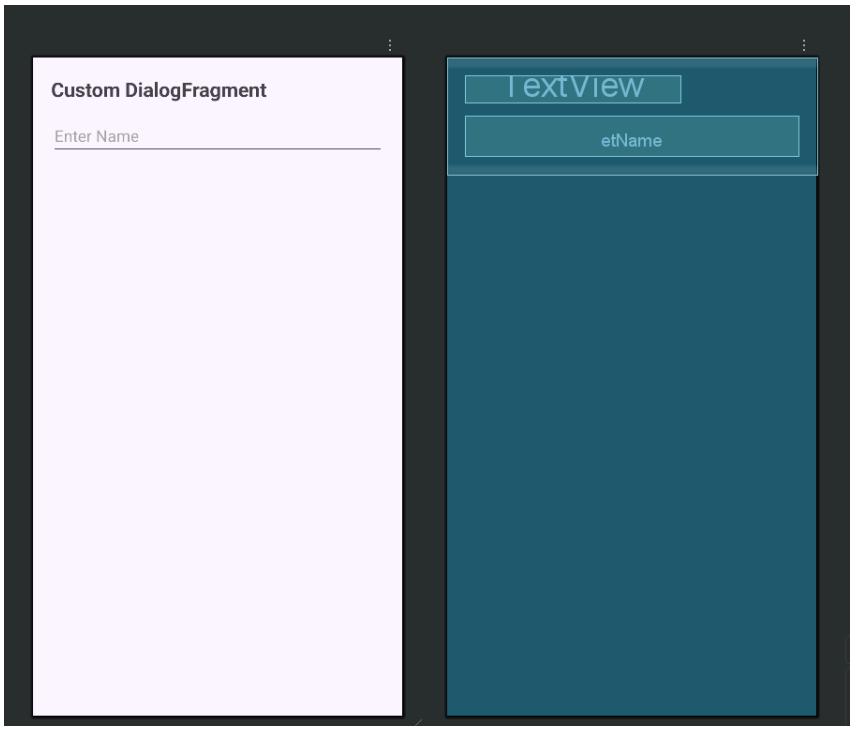
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

Create Custom Dialog Layout.

Create:

res→ layout → New-> XML-> New Layout XML File → dialog_custom



```xml
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="wrap_content"

   android:orientation="vertical"

   android:padding="20dp">

   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Custom DialogFragment"

       android:textSize="22sp"

       android:textStyle="bold"/>

   <EditText

       android:id="@+id/etName"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:hint="Enter Name"

       android:layout_marginTop="15dp"/>

</LinearLayout>
```

Create Dialog Fragment Class.
Right Click: java package → New → Java Class

Class Name: CustomDialogFragment.java

```
package com.example.dialogdemo;



import android.app.AlertDialog;

import android.app.Dialog;

import android.os.Bundle;

import android.view.LayoutInflater;

import android.view.View;

import android.widget.EditText;

import android.widget.Toast;

import androidx.annotation.NonNull;

import androidx.fragment.app.DialogFragment;


public class CustomDialogFragment extends DialogFragment {


   EditText etName;

   @NonNull

   @Override

   public Dialog onCreateDialog(Bundle savedInstanceState) {


       AlertDialog.Builder builder = new AlertDialog.Builder(getActivity());

       LayoutInflater inflater = requireActivity().getLayoutInflater();

       View view = inflater.inflate(R.layout.dialog_custom, null);
```

```
        etName = view.findViewById(R.id.etName);

        builder.setView(view)

                .setTitle("Custom Dialog")

                .setPositiveButton("Submit", (dialog, which) -> {

                    String name = etName.getText().toString();

                    Toast.makeText(getActivity(),

                            "Entered: " + name,

                            Toast.LENGTH_SHORT).show();

                })

                .setNegativeButton("Cancel", (dialog, which) ->
dialog.dismiss());

        return builder.create();

    }

}
```

FileName: MainActivity.java

```
package com.example.dialogdemo;
```

```


import android.os.Bundle;

import android.widget.Button;

import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;

import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {



   Button btnAlert, btnCustom;

   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);



       btnAlert = findViewById(R.id.btnAlert);

       btnCustom = findViewById(R.id.btnCustom);



       // AlertDialog Example

       btnAlert.setOnClickListener(v -> {

           AlertDialog.Builder builder = new AlertDialog.Builder(this);

           builder.setTitle("Delete File")

                   .setMessage("Are you sure you want to delete?")

                   .setPositiveButton("YES", (dialog, which) -> {

                       Toast.makeText(this,

                               "File Deleted",

                               Toast.LENGTH_SHORT).show();



                   })

                   .setNegativeButton("NO", (dialog, which) ->
dialog.dismiss());
```

```
            builder.show();

        });

        // Custom Dialog Example

        btnCustom.setOnClickListener(v -> {

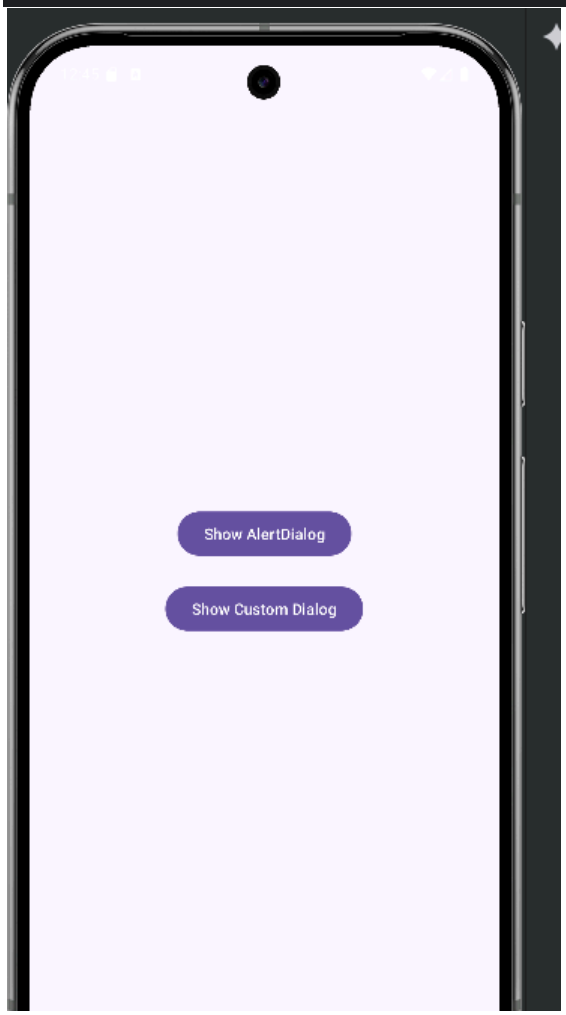
            CustomDialogFragment dialog = new CustomDialogFragment();

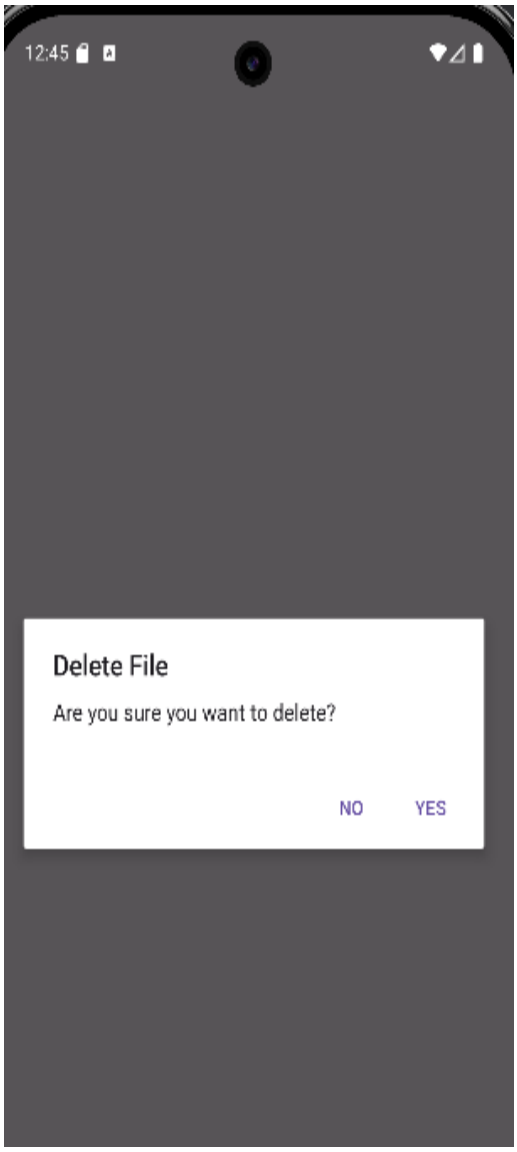
            dialog.show(getSupportFragmentManager(), "CustomDialog");

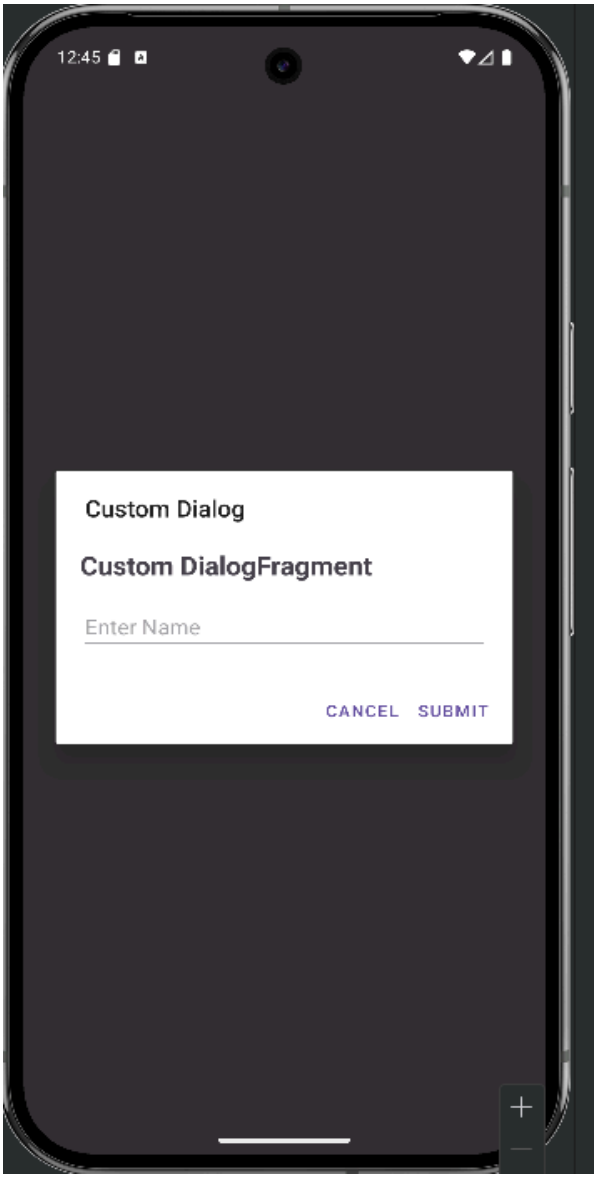
        });

    }

}
```




12:45
Delete File
Are you sure you want to delete?
NO
YES


12:45
Custom Dialog
Custom DialogFragment
Enter Name
CANCEL
SUBMIT